Practical Data Stewardship for Salmon Biologists

A Blueprint for Domain-Specific Best Practices

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Fisheries management and salmon conservation depend on integrating diverse datasets collected across jurisdictions, disciplines, and life-history stages. In practice, these data are often fragmented, inconsistently documented, and difficult to reuse, making synthesis and routine updates costly. Data stewardship—the social and technical work of documenting, governing, standardizing, and publishing data products—can reduce these barriers when supported by clear roles, shared practices, and fit-for-purpose infrastructure. We present seven practices for salmon data stewardship, grounded in FAIR and CARE principles, and illustrate them with a cross-jurisdictional sockeye productivity synthesis case study spanning Washington, British Columbia, and Alaska (Peterman & Dorner, 2012). We provide practical guidance and examples for biologists taking on stewardship responsibilities, including a brief worked example (SDIS) showing how the practices can be implemented as an end-to-end, versioned workflow. While motivated by salmon, the practices are intended to be transferable to other data-rich fisheries domains.

The Data Stewardship Challenge

Effective fisheries management relies on robust, accessible data on abundance, health, and environmental conditions integrated over large spatial areas and fine temporal scales. Yet across many fisheries, data remain fragmented, inconsistently measured, and incomplete across time, space, and life history stages, which limits their value for research, hypothesis testing, and decision-making (Atkinson et al., 2025; Diack et al., 2024; Inman et al., 2021; Marmorek

et al., 2011; Price & Moore, 2026). Collections are often managed by diverse agencies and organizations in isolation, undermining timely, integrated analyses necessary for adaptive management. The absence of organizational support and dedicated roles for data stewardship further compounds this problem, with critical tasks often treated as ad hoc responsibilities rather than core management functions. These issues limit the utility of these data for research and management.

Fragmented Landscapes

For salmon in particular, the integration and mobilization of salmon data mirrors the complexity of their life cycle, life histories, and migrations across ecological regions (Groot & Margolis, 1991) and jurisdictional boundaries. Salmon biologists routinely collect information managed by diverse agencies and organizations, often isolated within their operational context and without a focus on interoperability. While localized successes in data coordination exist—particularly within regional fisheries management offices and treaty commissions (*Fraser River Panel Data Application*, 2025)—salmon data integrated across agencies for each phase of the salmon life cycle is uncommon and costly for most programs. Even within organizations, data can be siloed by data type with freshwater, estuary, open-ocean, and commercial fishery data each housed in separate data systems with limited ability to easily re-connect the data through shared identifiers. The result is a highly fragmented data landscape that spans agencies, organizations, and monitoring programs, making integration especially difficult. This lack of coordinated stewardship has become a significant impediment for adaptive salmon management and conservation, preventing the full use of existing data to inform timely and effective decisions.

This fragmentation is compounded by the number of scientific disciplines involved in piecing together a comprehensive understanding of individual salmon stocks. Geneticists, oceanographers, freshwater ecologists, stock assessment biologists, and fisheries managers are just a sample of disciplines that contribute data to piece together a complete life history for a salmon population. Each uses their own domain-specific conventions and workflows, standards, and metadata. This tangle of disciplinary and organizational fragmentation slows integration, hinders reproducible synthesis and reanalysis, and delays analyses that could otherwise inform time-sensitive management decisions, conservation actions, or restoration plans. Large volumes of data collected through long-term monitoring programs hold tremendous value, especially for secondary users—but are often inaccessible due to a lack of time, resources, and incentives for data producers to publish them (LINDENMAYER et al., 2012). When critical datasets are hard to find, access, or interpret, biologists and analysts may spend substantial time reconstructing or harmonizing them. This can reduce transparency, increase the risk of errors, and delay conservation or management responses. Modernizing these systems will require coordinated communities of practices, shared practices, and shifting collective expectations and culture towards shared international data standards and stewardship practices that accommodate the full disciplinary and geographic diversity of salmon science.

Coordinated Data Stewardship

Data stewardship offers a practical way to address the growing challenges in salmon management arising from escalating environmental uncertainties due to climate change, which demand rapid, regionally integrated, and robust data Ward et al. (2025). Here, we use *data stewardship* to mean the social and technical work of documenting, governing, standardizing, and publishing data products so they can be responsibly reused over time. The mismatch between fragmented data systems and fixed administrative and jurisdictional boundaries creates an urgent need for interoperable, dynamic, multi-scale data stewardship that can adapt to shifting ecological and management priorities. Despite the scale and importance of these datasets, biologists who collect and manage salmon data are often expected to act as de facto data stewards without training, guidance, incentives, organizational support, or access to community-agreed best practices. Tasks such as documenting methods, aligning terminology, formatting for data sharing, and publishing data are typically performed off the side of a biologist's desk. Done well, data stewardship is skilled, labor-intensive work—combining domain expertise, information science, and relationship-building—and it requires dedicated time and institutional support rather than ad hoc effort. A lack of organizational support (Diack et al., 2024), training (Volk et al., 2014), and dedicated roles for data management further relegate critical data stewardship tasks to an *ad hoc* status. Deferred documentation, governance, and standardization tasks accumulate as data technical debt—work left undone that compounds over time, making integration and reanalysis progressively slower and more error-prone. The absence of clear roles, standards, and community-endorsed practices leaves even motivated scientists unsure how to structure their data for future use. As a result, data stewardship is inconsistent and reactive, and data integration remains a major bottleneck to adaptive management and ecosystem-scale learning.

Due to a lack of practical resources, many salmon biologists struggle to align their data with community-agreed principles such as FAIR (Findable, Accessible, Interoperable, and Reusable) (Wilkinson et al., 2016) and Indigenous Data Sovereignty frameworks like the CARE principles (Collective Benefit, Authority to Control, Responsibility, and Ethics) (Carroll et al., 2019; Jennings et al., 2023). CARE principles are particularly salient in salmon data contexts given the sociocultural importance of salmon to First Nations, Tribes, and Indigenous communities throughout the North Pacific and North Atlantic regions (Earth Economics, 2021; Ween & Colombi, 2013). While both FAIR and CARE present laudable goals with respect to data stewardship, these sets of principles can sometimes be at odds with each other and implementing both presents an inherent challenge (Tattersall et al., 2025). Nonetheless, several of the best practices we propose can help clarify shared objectives and increase transparency and trust that is essential for integrating both FAIR and CARE principles. Without clear support and guidance, well-intentioned practitioners are left with ad hoc approaches that limit reuse and interoperability. This gap can only be bridged by equipping both data producers and stewards with tools, support, and organizational backing to publish interoperable, machine-readable metadata and datasets in alignment with shared principles.

Framework for Action

In this paper, we provide a framework for translating data stewardship principles into actionable practices, examples, and workflows to help salmon biologists improve the usability, traceability, and long-term impact of their data. We offer seven practices for salmon data stewardship, with the Best Practices table designed as the quick-start entry point: the left column defines each practice and the right column lists concrete next steps. We then use a retrospective, real-world case study of cross-jurisdictional sockeye productivity synthesis (Peterman & Dorner, 2012) to illustrate how the absence of shared practices made integration time-consuming and keeps annual updates costly and difficult to this day. To support adoption, we provide real-world examples from multiple organizations (Appendix 1), a training roadmap for biologists taking on stewardship responsibilities (Appendix 2), and a getting-started checklist (Appendix 3). For organizations that require lifecycle framing, the same seven practices can be applied across planning, processing, preservation, access, and governance. Although salmon provide the worked example, the approach is intended as a transferable blueprint: other scientific communities can substitute their own vocabularies, repositories, governance structures, and incentives to turn data stewardship principles into practical applications. We also include a short worked example—the Salmon Data Integration System (SDIS)—to show how the practices can be implemented as an end-to-end, versioned workflow in an operational salmon data program.

> **ℹ** Key terms (quick primer)
>
> - **Schema / data model:** a structured description of what fields exist and how they relate.
> - **Controlled vocabulary:** an agreed set of permitted terms (often with stable identifiers) to reduce ambiguity.
> - **Ontology:** a more formal model of concepts and relationships that supports mapping across datasets and systems.
> - **Data package:** data + metadata + validation context, bundled as a reusable release (i.e., a contract for reuse, not just a file).
> - **Machine-actionable:** information is structured enough that software can reliably find, validate, and use it (not just read it).

A Data Management Plan (DMP) explains how data will be collected, documented, stored, and preserved across a project. A Data Product Governance Charter defines decision rights, stewardship responsibilities, and quality expectations for an ongoing data product. A Data Sharing Agreement specifies who can access which data, for what purposes, and under what legal, ethical, and governance conditions.

A coordinated approach to stewarding salmon data should follow established open science standards and principles (Carroll et al., 2020; Johnson & Stap, 2024; Wilkinson et al., 2016), tailored specifically within the context of salmon research and management. Our practices build upon and extend existing data and metadata standards and infrastructure including Darwin

Core, OBIS, schema.org, and OBO Foundry ontologies, ensuring compatibility with broader biodiversity informatics infrastructure while avoiding reinventing foundational frameworks. Achieving meaningful interoperability of data among programs, organizations and scientific domains demands both breadth and depth. **Broad interoperability** integrates diverse scientific domains, systems, and formats, requiring structured, machine-readable data and metadata published with clear, machine-readable access conditions to support appropriate discovery and reuse. **Deep interoperability** demands precise definitions of domain-specific terms and methods, ensuring data remains meaningful and usable across contexts.

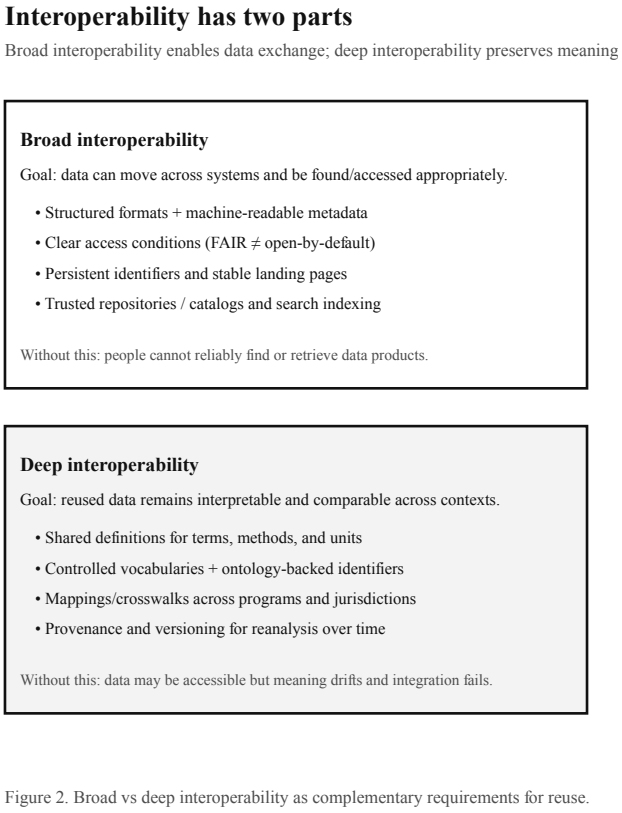

Figure 1: Broad interoperability enables exchange and discovery; deep interoperability preserves meaning for integration and reanalysis.

Salmon data stewards can improve outcomes for salmon by coordinating across boundaries to develop a shared foundation of data stewardship practices. To address these foundational challenges, it is important to establish clear roles and practices that span both the data lifecycle and the salmon life cycle—from collection and documentation through integration, long-term preservation, and reuse, and from gravel to estuary and the high seas back to gravel.

Defining Data Stewardship in Salmon Science

Data stewardship encompasses the coordinated practices, roles, and responsibilities necessary to effectively manage, share, and reuse data throughout its lifecycle (NOAA, 2007; Peng et al., 2018; Plotkin, 2014). It includes ensuring data quality, compliance with agreed-upon standards, and the establishment of clear governance to guide data collection, documentation, integration, and preservation. However, given the complexity of the salmon data landscape, stewardship goes beyond mere technical data management to include actively facilitating collaboration, communication, and consensus-building among data producers and users across multiple organizations and jurisdictions.

Effective salmon data stewards serve as boundary spanners bridging agencies, Nations, and disciplines serving as community coordinators, convening diverse stakeholders to build sustained communities of practice. This boundary-spanning role is particularly critical in transboundary contexts where data integration requires navigating complex jurisdictional and cultural boundaries (Ward et al., 2025). These responsibilities demand specialized expertise and sustained effort; without dedicated capacity, standards and workflows tend to degrade, and technical debt accumulates. By facilitating communication, translating between different organizational cultures and technical systems, and maintaining long-term relationships, data stewards create the social infrastructure necessary for effective cross-boundary data collaboration.

Box 1: Common Functions of Salmon Data Stewards

Effective salmon data stewards often perform several common functions:

- **Technical oversight**: Ensuring metadata completeness, adherence to standardized terminologies and vocabularies, and robust quality assurance protocols.
- **Social and organizational facilitation**: Leading stakeholder engagement, capacity-building activities, and negotiation of data access and sharing agreements, including addressing First Nations, Tribes, and Indigenous Peoples' rights and interests in data governance.
- **Organizational advocacy**: Championing the organizational recognition of data stewardship roles, promoting sustained investment and dedicated resources for data management infrastructure and practices.
- **Implementation and adoption facilitation**: Actively promoting data use and ensuring that standards and practices remain practical and relevant by maintaining close contact with real-world applications. This includes monitoring data utilization, gathering feedback from users, and iteratively refining standards based on actual implementation challenges to prevent theoretical approaches that fail in practice.

Data stewards can implement FAIR and CARE principles through concrete technical and

governance mechanisms they control, such as documenting consent constraints and access levels in metadata, using controlled vocabularies to ensure consistent terminology, and establishing repository roles that enforce data sovereignty requirements. FAIR is sometimes interpreted as "open by default," but it does not require unrestricted openness; rather, it requires that data and metadata be discoverable and accessible under explicit, machine-readable conditions (GO FAIR Initiative, n.d.; Landi et al., 2020). CARE's Authority to Control can therefore place legitimate constraints on access and reuse, particularly when data are held or governed by First Nations, Tribes, and Indigenous Peoples. In these cases, stewardship often means publishing rich, standards-aligned metadata while implementing governed access, consent-aware reuse conditions, and culturally appropriate protocols for the data themselves. For example, stewards can document consent constraints in metadata fields and enforce access restrictions via repository user roles, ensuring that Indigenous data sovereignty is respected while maintaining data discoverability and appropriate reuse (Local Contexts, 2025; Montenegro, 2019). This governance approach is particularly critical for sensitive data such as Traditional Knowledge and sensitive habitat locations, where stewardship practices must balance discoverability and reuse objectives with appropriate access controls and cultural protocols (Carroll et al., 2021; Chapman, 2020).

Data stewards play a critical role bridging the gap between biologists and Information Technology (IT) staff by translating data needs into application or data system features. A user-centred design approach to salmon data stewardship is critical and focuses on creating tools that align with biologists' needs. When data management is separated from biologists, accountability weakens, and quality issues go unnoticed. While IT expertise is essential for infrastructure and security, effective data system design works best when IT enables secure, well-supported self-serve infrastructure while domain teams define stewardship requirements and governance. The Data Steward, serving as a translator between IT and biologists, enables biologists to engage independently with data systems, strengthening feedback loops and accountability for research and management.

Dedicated stewardship roles empower salmon biologists to bridge disciplinary divides and jurisdictional barriers, transforming fragmented datasets into cohesive, interoperable resources. By proactively defining, implementing, and maintaining data standards and workflows, salmon data stewards create conditions for timely, accurate, and reproducible analyses. Such stewardship positions salmon biologists to better inform adaptive management decisions, ultimately strengthening salmon conservation and resilience.

Updating Sockeye Productivity Synthesis: A Case Study for What Agencies Could Do Now

This case study revisits a comprehensive Sockeye Salmon Oncorhynchus nerka productivity dataset assembled from several agency sources by academic researchers (Peterman & Dorner, 2012). We reflect not on the significant work the research team accomplished, but rather on the preventable organizational and technical barriers that impeded their work—and continue to

burden data compilation updates and reuse efforts today. Their study examined productivity trends across 64 Sockeye Salmon stocks spanning Washington, British Columbia (B.C.), and Alaska and found common trends in declining productivity across the North American range of Sockeye Salmon (Peterman & Dorner, 2012). This work was important for sockeye conservation and management because it synthesized productivity information across jurisdictions at a scale that is difficult to achieve without coordinated, cross-program data integration. Attempting to replicate or build upon this analysis today can be time-consuming and error-prone due to fragmented data sources, inconsistent formats, and lack of standardized practices among the key organizations involved: the Washington Department of Fish and Wildlife (WDFW), Fisheries and Oceans Canada (DFO), and the Alaska Department of Fish and Game (ADF&G).

Each section below highlights a key challenge faced by the team and proposes practical steps based on our best-practices framework that data-holding agencies could use to enable easier integration, validation, and updating of salmon datasets across jurisdictions and decades. This case study illustrates how implementing the foundational concepts and practical recommendations outlined in this paper can transform data stewardship practices within these organizations. By doing so, they can significantly enhance data accessibility, quality, and interoperability—ultimately enabling more efficient and accurate analyses that support salmon conservation and management.

Challenge 1: Interpreting the Data — What do these numbers actually mean?

Peterman's team frequently worked with datasets that lacked basic contextual information. Fields such as "year," "return," or "age class" were often undefined or inconsistently used. For example, some datasets recorded returns by calendar year while others used brood year, and few included metadata to clarify the distinction. In many cases, the team had to reconstruct metadata by back-checking against reports or simulating assumptions (e.g., about age structure) to interpret the data correctly.

Remedies:

- **Best Practice 3: Make Data, People, Projects, and Outputs Discoverable, Linked and Citable with Persistent Identifiers (PIDs).** Assigning PIDs such as digital object identifiers (DOIs) to protocols, methods, and people (via ORCIDs) and linking them together using data stores and catalogues links data to its provenance and ensures that methods, context, and interpretation decisions are traceable.
- **Best Practice 4. Use Shared Data Models, Vocabularies and Metadata to Enable Integration.** To prevent this kind of ambiguity, agencies can now adopt internationally recognized metadata schemas such as ISO 19115 or Ecological Metadata Language, data models (Darwin Core Data Package) to model age and age type data concepts, and use controlled vocabularies to restrict the permissible values in the age field to calendar year, brood year, or otherwise.

Challenge 2: Accessing and Using the Data — Where is it stored, and how do I get it?

The Peterman dataset was compiled from multiple files scattered across email inboxes, regional offices, and grey literature. Data were stored in inconsistent formats, lacked clear versioning, and were difficult to discover outside of specific research networks. Even today, no application programming interface (API) or structured access mechanism exists to update or query the data programmatically. As a result, researchers hoping to build on the dataset may have to start from scratch.

Remedies:

- **Best Practice 2: Reuse Proven Infrastructure to Save Time and Increase Interoperability**
  Rather than developing bespoke catalogues or repositories, agencies should prefer established, community-adopted repositories and catalogues that fit their data type and governance requirements (e.g., OBIS for biodiversity occurrence-style datasets; Zenodo or KNB for general research outputs). Established platforms can improve discoverability and reuse when paired with good metadata and stewardship.
- **Best Practice 5: Store and Analyze Data in Ways That Others Can Easily Access, Use, and Trust**
  Agencies can use repositories or catalogues that make data discoverable using PIDs and support appropriate access mechanisms (including governed access and, where useful, programmatic access such as APIs).

Challenge 3: Sustaining the Dataset — Who is responsible, and why should I contribute?

Once Peterman and his team completed their analysis, no formal plan existed for sustaining or updating the dataset. Responsibility for ongoing maintenance fell informally to former students and collaborators. Despite its national and international relevance, the dataset was never adopted by an agency as a living product. Moreover, the original data contributors often lacked incentives, support, or recognition for their efforts—conditions that persist in many data environments today.

Remedies:

- **Best Practice 1: Make Data Governance Explicit to Support Trust and Reuse**
  Agencies should define roles, responsibilities, and decision-making processes through formal governance mechanisms such as data product charters. Use a Data Management Plan with a responsibility matrix such as "responsible, approver, consulted, informed" (RACI) to clarify governance, assign maintenance responsibility, and ensure continuity across staff turnover and organizational change.

- **Best Practice 6: Incentivize and Track Data Sharing and Reuse** Visibility, credit, and metrics are critical for motivating data sharing. Agencies can embed citation guidance in metadata and track dataset reuse through COUNTER-compliant dashboards or DataCite APIs.
- **Best Practice 7: Build Community Through Co-Development and Mutual Benefit** Effective data stewardship requires collaboration between biologists, First Nations, Tribes, Indigenous communities, managers, and data professionals. Participatory design ensures that systems and standards meet user needs and are adopted over time. *Practical application:* Facilitate cross-jurisdictional working groups to co-develop data standards and align on shared outcomes for priority datasets.

While the barriers encountered in compiling, interpreting, and maintaining the data made the work considerably more difficult than necessary and hinder efforts by others seeking to extend or build upon this work, they remain instructive. These challenges are not unique to Peterman's team—they reflect systemic gaps in data governance, documentation, infrastructure, and incentives. By adopting the seven best practices detailed in the Best Practices table, organizations and researchers can improve the reusability of legacy datasets, easing collaboration and supporting more timely synthesis across the salmon research and management community.

The challenges and solutions demonstrated in this salmon case study generalize across fisheries and environmental monitoring domains. Cross-jurisdictional data harmonization, quality assurance and control patterns, standardized metadata requirements, and long-term archiving strategies are universal needs that extend far beyond salmon science. Similar barriers and solutions apply to trawl survey data integration, invertebrate monitoring programs, and water quality datasets that span multiple agencies and jurisdictions.

Table 1: Best practices for salmon data stewardship with quick-start next steps

Best Practice	Next steps (examples)
1. Make Data Governance Foundational and Explicit to Establish and Sustain Trust and Reuse. Establishing clear governance structures promotes quality, accountability, and alignment with FAIR and CARE principles enabling trust and long-term stewardship across multi-organization projects.	• Document roles and responsibilities using a Data Product Governance Charter and structured responsibility frameworks. • Integrate CARE principles to respect First Nations, Tribes, and Indigenous data rights. • Form a governance or oversight committee to review data standards, timelines, and agreements.

Best Practice	Next steps (examples)
2. Reuse Proven Infrastructure to Save Time and Increase Interoperability. Leveraging existing platforms and technologies by building on and extending them rather than building bespoke solutions reduces costs and improves long-term interoperability and sustainability.	• Use domain-specific repositories. • Publish and archive data with established repositories (e.g., KNB, Zenodo) matched to your data type and governance context.
3. Make Data, People, Projects, and Outputs Discoverable, Linked and Citable with Persistent Identifiers. Persistent identifiers (PIDs) connect data with researchers, organizations, and outputs—supporting data citation, reuse, and automated attribution.	• Encourage ORCID use for researchers. • Use ROR IDs for organizations. • Assign DOIs via DataCite for data packages. • Embed DOIs in dashboards and metadata.
4. Use Shared Data Models, Vocabularies and Metadata to Enable Integration. Common vocabularies, metadata standards, and ontologies support integration across systems and preserve semantic meaning.	• Adopt ISO 19115, EML, or DataCite metadata standards. • Re-use terms defined in Salmon Domain Ontology. • Model datasets using the Darwin Core Data Package model.
5. Store and Analyze Data in Ways That Others Can Easily Access, Use and Trust. Structured and accessible data formats ease reusability and integration with analytical tools, while programmatic workflows (e.g., R, Python) increase reproducibility and trust.	• Provide programmatic access (e.g., APIs) where appropriate. • Archive in trusted repositories (e.g., GBIF, FRDR, USGS). • Use scripts to wrangle, transform, and analyze data reproducibly. • Use GitHub for transparent collaboration and GitHub/Zenodo for DOI assignment and preservation.
6. Incentivize and Track Data Sharing and Reuse. Recognizing data contributors and tracking reuse promotes a culture of sharing and supports professional recognition.	• License data with CC-BY 4.0. • Include citation text and visible credit fields. • Use COUNTER metrics and DataCite APIs to monitor reuse. • Encourage dataset citation in references.
7. Build Community Through Co-Development and Mutual Benefit. Engaging users early ensures tools and standards meet real-world needs and enhances long-term stewardship.	• Participate in RDA Salmon Interest Group. • Facilitate workshops for metadata and vocabulary alignment. • Support community-engaged research with tangible benefits.

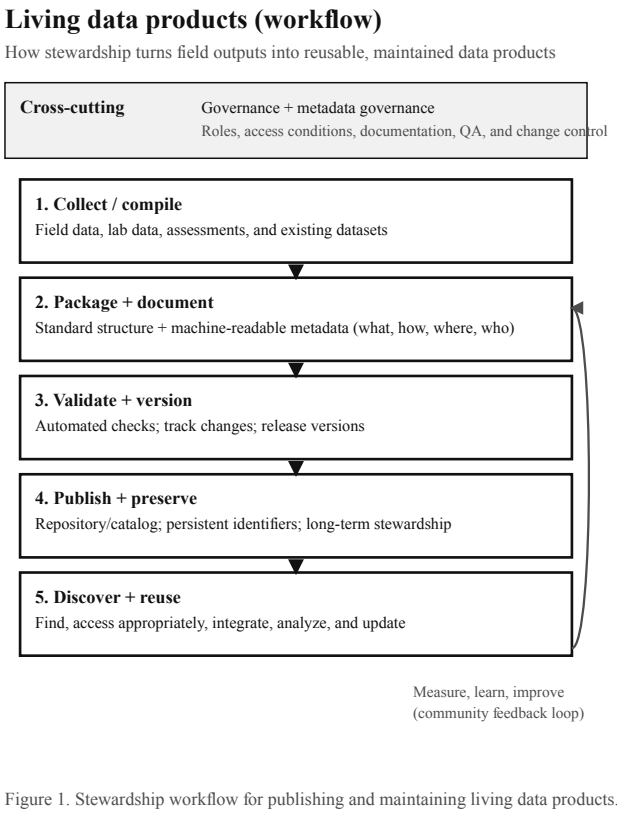

Figure 2: Stewardship workflow for publishing and maintaining living data products.

Box: Salmon Data Integration System (SDIS) — a worked example

SDIS (Salmon Data Integration System) is a worked example of how the seven best practices can be implemented together in a salmon monitoring context. It is not a prescribed software stack; it is a transferable pattern: make governance and access conditions explicit, package data with standardized machine-readable metadata, use controlled terminology and persistent identifiers, validate automatically, and publish versioned, citable releases.
Core SDIS components (illustrative):

- **metasalmon (R)** (DFO Pacific Science, 2026b) (https://dfo-pacific-science.github.io/metasalmon/index.html)
- **Salmon Data Package specification** (DFO Pacific Science, 2026c) (https://github.com/dfo-pacific-science/smn-data-pkg/blob/main/SPECIFICATION.md)
- **Salmon Domain Ontology** (canonical interoperability layer for shared salmon concepts)
- **DFO Salmon Ontology** (DFO Pacific Science, 2026a) (https://dfo-pacific-science.github.io/dfo-salmon-ontology/) (organization-specific ontology and controlled vocabularies aligned to the broader Salmon Domain Ontology layer)
- **Salmon Data GPT** (DFO Pacific Science, 2026d) (https://github.com/dfo-pacific-science/smn-gpt)

Other programs can implement the same stewardship pattern using different tools. Where provenance is important, relationships among datasets, methods, and releases can be expressed using standard conceptual models (e.g., PROV-DM) (W3C, 2013).

In SDIS, governance, metadata, packaging, validation, and semantics are treated as one operational workflow to support living datasets that remain interpretable and reusable across time, teams, and institutions.

For teams that use lifecycle framing, these practices operate across the full data lifecycle, with metadata governance treated as a continuous cross-cutting function rather than a separate step.

Conclusion

Salmon biologists and data stewards across the globe have generated extensive datasets on salmon abundance, environmental conditions, and biological characteristics. When integrated, these data become valuable assets, a fact powerfully demonstrated by studies such as (Peterman & Dorner, 2012). However, as noted by reports to the Cohen Commission investigating the decline of Fraser River sockeye salmon (Marmorek et al., 2011), these data are often incomplete, inconsistently collected, and fragmented across organizations and jurisdictions. Integrating

across such diverse sources can be done but requires effort that is often not accounted for in smaller-scale studies or perpetually underestimated in long-term ongoing data projects. This fragmentation hinders a deeper understanding of the drivers of change across salmon life stages and regions, and limits the effectiveness of management decisions, particularly concerning in the face of anthropic climate change and biodiversity loss.

We argue that many persistent bottlenecks in salmon analysis and management are fundamentally data problems, and that coordinated data stewardship offers a high-leverage, near-term opportunity for improvement. By adopting shared best practices in data governance, metadata standardization, persistent identification, infrastructure reuse, and community co-development we can meaningfully improve the transparency, reusability, and interoperability of salmon data. A coordinated, future-oriented data stewardship strategy can leverage the full potential of data in science and management. The case study presented in this paper—drawn from one of the Pacific Region's most influential salmon survival syntheses (Peterman & Dorner, 2012)—illustrates how technical and social data management gaps significantly slowed efforts to answer pressing questions. If some of the best practices we propose had been adopted earlier, time and resources devoted to integration and updates might have been reduced. The case study offers a cautionary tale and a practical set of lessons. Because data practices underpin nearly every analysis and decision, strengthening data stewardship is a practical way salmon programs can improve outcomes within their control.

The emergence of the data stewardship role (Plotkin, 2014) represents an important organizational shift needed to realize this vision. Historically, the work of managing, documenting, and maintaining data has been diffuse and undervalued—often falling to biologists without support, training, or recognition. As the volume and complexity of scientific data grow, so too does the need for clearly defined data stewardship responsibilities embedded within research teams and organizations. Training biologists in the principles and practices of data stewardship—while also supporting dedicated professionals who specialize in this work—is essential to sustaining trustworthy, reusable, and interoperable salmon data systems.

Realizing this vision requires concrete organizational commitments. For example, organizations can formally appoint dedicated data stewards with clear roles, responsibilities, and reporting structures. Agencies can adopt centralized metadata repositories and establish compliance metrics to track progress toward FAIR and CARE principles. Key implementation steps include: (1) designating stewardship roles within existing organizational structures, (2) investing in metadata management infrastructure, (3) establishing data governance committees with cross-organization representation, and (4) developing performance indicators that measure data discoverability, interoperability, and reuse. These organizational changes ensure that data stewardship becomes embedded in organizational culture rather than remaining an ad hoc responsibility.

The visionary future state is one where salmon researchers and stewards—across agencies, Indigenous Nations, academic laboratories, and community groups—can easily access and contribute to well-documented, versioned, and machine-readable datasets. In this future, field biologists, Indigenous guardians, modellers, and policymakers interact with a living

knowledge system—one that is flexible, easy to implement, and rooted in principles of FAIRness and Indigenous Data Sovereignty. Metadata standards, controlled vocabularies, and shared governance frameworks are not afterthoughts but integral to the culture of data collection and use. Scientists receive credit for publishing high-quality data, and users trust the provenance and structure of the datasets they rely on to make critical management decisions.

Realizing this vision will require investment in both people and systems. Key to this transformation is the emergence of the data steward as a professional role: a hybrid expert who understands operational field biology, information science, governance protocols, and community needs. As highlighted by Roche et al. (2020), institutionalizing data stewardship roles ensures long-term capacity for data governance, quality control, and interoperability—functions that are often neglected or left to informal actors. We should not only train new data stewards but also support and upskill biologists to take on stewardship responsibilities in collaborative, interdisciplinary settings. This is essential to address the data technical debt of unmanaged data and to modernize research practices in line with open science norms. By embedding these practices into the everyday work of data generation, documentation, publication, and reuse, we can move salmon science decisively into the era of data-intensive discovery.

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Data Availability

This manuscript is a synthesis and does not report a newly generated primary dataset. Datasets referenced as examples are available through the cited repositories, landing pages, and DOIs in the text and appendices.

Code Availability

No new software was developed specifically for this manuscript. Code resources discussed as examples are available at the cited public repositories and archival records.

Conflicts of Interest

None declared.

Ethics Statement

This manuscript does not involve primary research with human or animal subjects. All cited research was conducted in accordance with applicable ethical guidelines and legal requirements of the jurisdictions in which the studies were performed.

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Appendices

Appendix 1. Real-world Example Applications of the Best Practices

Here we provide detailed descriptions of the seven best practices for salmon data stewardship, along with practical applications and real-world examples. This is not an exhaustive list, but rather a starting point for salmon biologists and data stewards to implement effective data stewardship practices in their work based on examples from the salmon research and management community.

1. Make Data Governance Explicit to Support Trust and Reuse

Clear governance defines roles, responsibilities, and procedures ensuring data quality, long-term maintenance, accountability, and compliance with community principles such as FAIR and CARE. Effective governance fosters trust, and facilitate data sharing, reduces ambiguity regarding decision making, and is critical for coordinating both technical and sociocultural aspects of data stewardship.

In collaborative international or multi-organizational settings, establishing governance at the outset of a project is crucial for aligning diverse groups, including biologists, data managers, Indigenous communities, policymakers, and other participants. Early governance planning should establish clear, collaborative frameworks that respect each group's expertise and needs from the beginning.

Practical Applications:

1.1 Document roles and responsibilities clearly at project start using a Project or Data Product Governance Charter and structured frameworks (e.g., DACI or RACI charts) that relate to organizational data policies.

- Example of a Data Management Plan from the California Department of Water Resources
- Data Management Plan Templates from DMPTool, and NOAA Data Management Handbook

1.2 Integrate CARE principles to ensure ethical governance and respect Indigenous data rights.

- Northwest Indian Fisheries Commission use password protected website to host all the WDFW and tribal data in a one-stop shopping website for co-managers to pull data they need for decision-making process. https://fisheriesservices.nwifc.org/

1.3 Create a governance or oversight committee for regular data practice reviews and decision making regarding data structures, timelines, data sharing agreements and interoperability protocols

- Pacific Salmon Commission has formed a Technical Committee on Data Sharing including both US and Canadian data contributors. https://www.psc.org/membership-lists/

2. Reuse Proven Infrastructure to Save Time and Increase Interoperability

Building custom solutions should be avoided where possible. Maximizing existing platforms and technologies reduces costs, accelerates implementation, and increases data interoperability. Building modular, interoperable systems grounded in proven technologies ensures sustainable long-term stewardship.

Practical Applications:

2.1 Use the Ocean Biodiversity Information System or the Global Biodiversity Information Facility to standardize and host your data

2.2 Use free data catalogue services such as the Knowledge Network for Biocomplexity (KNB) or Zenodo

- The Pacific Salmon Foundation's spawner surveys dataset on Zenodo (Carturan and Peacock 2025) received more views within weeks than a analogous dataset in the Salmon Data Library (Pacific Salmon Foundation 2025) did over several years, illustrating that leveraging established public data infrastructures, rather than developing institution-specific ones, can substantially increase discoverability.

3. Make Data, People, Projects, and Outputs Discoverable, Linked and Citable with PIDs

Persistent identifiers (PIDs), including Digital Object Identifiers (DOI) are essential for tracking the provenance and reuse of data, and linking data, protocols, organizations and people. They allow for consistent referencing, integration across systems, and automated credit via data citation.

Practical Applications:

3.1 Encourage researchers to register for an Open Researcher and Contributor ID (ORCID) and include ORCIDs in metadata records and submission forms

3.2 Register your organization with the Research Organization Registry (ROR) and use ROR IDs to identify institutions involved in salmon science.

- Several salmon data holding institutions are already registered with ROR. As a result, those organizations can track and demonstrate their scholarly impact from data publications: DataCite Commons: Pacific Salmon Foundation

3.3 Assign DOIs to data packages, protocols, and reports using DataCite. Maintain version history for all metadata records and document the provenance of metadata creation, updates, and quality control processes to ensure accountability and traceability.

- The North Pacific Anadromous Fish Commission (NPAFC) assigns DOIs to IYS-related data packages which are served by a CKAN catalogue at https://data.npafc.org. The Commission also assigns DOIs to NPAFC Technical Reports and Bulletins.
- The Pacific Salmon Foundation's State of Salmon report (stateofsalmon.ca) has unique DOIs for each version, improving specificity and citability of each year's results.

3.4 Embed DOIs in dashboards, figures, and metadata so they persist in derivative products.

4. Use Shared Data Models, Vocabularies and Metadata to Enable Integration

Standardizing metadata and terminology ensures data can be interpreted correctly and integrated across systems. Controlled vocabularies, community ontologies, and structured metadata schemas allow data to retain its full semantic meaning.

Practical Applications:

4.1 Configure data catalogues and metadata intake tools to accept internationally recognized metadata schemas such as ISO 19115, Ecological Metadata Language (EML), or DataCite. Implement automated validation against schema requirements and manual review processes to ensure metadata completeness, accuracy, and consistency before publication.

- The Pacific Salmon Foundation's data portal asks contributors to provide metadata in ISO 19115 or other standard formats. marinedata.psf.ca, ensuring consistent metadata structure
- The NPAFC uses ISO 19115 metadata standard in their data catalogue https://data.npafc.org

4.2 Model datasets and databases using the Darwin Core Standard

- The Hakai Institute Juvenile Salmon Program publishes their data to OBIS using Darwin Core: Hakai Institute Juvenile Salmon Program
- The International Year of the Salmon High Seas Expeditions data mobilization efforts (Johnson & Stap, 2024) published their data to OBIS: https://www.gbif.org/dataset/search?project_id=IYS

- StreamNet's Data Exchange Standard (DES) provides a concrete example of collaboratively developed data exchange rules (structures, definitions, and business rules) for fish monitoring and related data (StreamNet, 2024).

4.3 Re-use or publish data terms that are shared online using a persistent identifier in a controlled vocabulary or ontology

- Treat the Salmon Domain Ontology as the canonical interoperability layer for shared salmon concepts across organizations.
- The DFO Salmon Ontology is one organization-specific ontology and controlled vocabulary set aligned to that broader layer, publishing stable identifiers and definitions to support semantic alignment and crosswalks: https://dfo-pacific-science.github.io/dfo-salmon-ontology/
- WDFW has definitions of all hatchery escapement data. Hatchery escapement reports | Washington Department of Fish & Wildlife
- Fish Passage Counts has defined metadata that can be used across OFDW and WDFW. https://www.fpc.org/111_sharedfiles/ColumbiaRiverBasinAdultFishPassageCountsMetadata.pdf

Best Practice 5: Store and Analyze Data in Ways That Others Can Easily Access, Use, and Trust

Making data easily accessible promotes its use in research and management, enabling integration with tools and applications. Ensuring accessible, persistent data storage requires more than file hosting (CoreTrustSeal, 2022). Data should be structured and well-documented, stored in repositories that support long-term preservation, and—where it materially improves reuse—made available via programmatic access such as APIs.

Practical Applications:

5.1 Provide direct data access via application programming interfaces (APIs) so users can retrieve, filter, and integrate data programmatically into analysis pipelines and decision-support tools.

- The Pacific States Marine Fisheries Commission makes its PIT Tag Information System data programmatically accessible via the PTAGIS API.
- The NPAFC CKAN catalogue exposes machine-actionable search and metadata endpoints (for example, package search for salmon datasets), enabling scripted discovery and harvesting.

5.2 Archive data in long-term, domain-relevant repositories that provide stable landing pages, machine-readable metadata, and explicit reuse conditions.

- NPAFC publishes salmon datasets with DOI-backed landing pages in its catalogue (for example, Genetic stock identification..., DOI: https://doi.org/10.21966/a59e-0t51).
- The Hakai Institute Juvenile Salmon Program provides biodiversity records through OBIS/GBIF infrastructure with stable dataset pages and downloadable resources: https://www.gbif.org/dataset/72de3af4-1572-4f2d-8006-2bfa2007065c.
- USGS ScienceBase hosts salmon-relevant geospatial products with persistent item pages and downloadable files (for example, State of the Salmon Range-wide Salmon Distribution).
- NOAA Fisheries maintains operational salmon data resources with annual updates (for example, U.S. Atlantic Salmon Assessment Committee Database).

5.3 Leverage the integration between GitHub and Zenodo to automate archiving and DOI assignment, ensuring long-term data preservation.

- The Pacific Salmon Foundation conducted an extensive cleaning of DFO's New Salmon Escapement Database System (NuSEDS) to maximize data recovery. The R code for this procedure is available on GitHub, while the resulting dataset is archived on Zenodo, with accompanying R Markdown documentation linked on both platforms. The code and dataset are updated in parallel with NuSEDS, and each dataset version is assigned a DOI to ensure proper citation and traceability.

6. Incentivize and Track Data Sharing and Reuse

The currency of research lies in recognition—credit, citations, and opportunities for collaboration or co-authorship. Promoting data sharing requires both cultural and technical infrastructure. The cultural infrastructure requires a shift towards viewing data publication as equal in importance to article publication. The infrastructure put in place needs to support the process of generating citation records that give credit to all First Nations, Tribes, agencies, and organizations. By recognizing contributions, tracking reuse, and supporting citation, data stewards can create a system where sharing is rewarded.

Practical Applications:

6.1 License data for reuse using liberal licenses

- All data accessible through the NPAFC data catalogue is licensed as Creative Commons Attribution 4.0 International

6.2 Provide recommended citation text and visible credit fields in metadata

- NPAFC dataset landing pages expose citable DOI metadata and a dedicated citation section alongside contributors and licensing details (example dataset: https://data.npafc.org/dataset/ca-cioos_2dac6609-3541-458c-88b1-5564599d9621?local=en).

- NOAA InPort records provide reusable citation text directly on metadata landing pages (example: AFSC/ABL: Karluk sockeye salmon scale time series).

6.3 Create summary dashboards that highlight reuse using COUNTER Code of Practice compliant metrics to track dataset views/downloads and DataCite APIs

- Use DataCite REST API endpoints to retrieve DOI metadata programmatically and feed organization-level tracking workflows.
- Combine repository-native metrics (views/downloads) with catalogue/API outputs (for example, NPAFC CKAN API results) to create repeatable monitoring reports.

6.4 Ensure that datasets are properly cited in journal articles using in text citations and the recommended citation in the articles list of references, not just in a Data Availability statement

- In late 2024, the NPAFC began citing data sets using in-text citations and the recommended citation in the list of references with the publication of NPAFC Bulletin 7 titled, *Highlights of the 2022 International Year of the Salmon Pan–Pacific Winter Expedition*.
- The cleaned NuSEDS dataset (see Section 5.3) was used in an analysis and cited in the associated publication (Atkinson et al., 2025) using the DOI and citation provided with the dataset on Zenodo.

6.5 Promote the view that well-documented data publications are primary research outputs and significant contributions to the field

- Treat dataset publication milestones (for example, DOI assignment, metadata completion, and versioned release) as reportable scholarly outputs in project and performance reporting.
- Encourage teams to include dataset DOIs in CVs, annual reports, and program dashboards alongside article outputs.

7. Build Community Through Co-Development and Mutual Benefit

Creating an infrastructure that standardizes and provides cross-border and cross-ecosystem data integration is only effective if there's community engagement. Standards and tools must be co-developed with their intended users using user-centred design principles (ISO 9241-210, 2019) to be effective. Engaging biologists, Indigenous stewards, and data managers ensures relevance, usability, and long-term participation.

Practical Applications:

7.1 Participate in salmon data focused communities such as the Research Data Alliance's Salmon Research and Monitoring Interest Group

7.2 Run participatory workshops for metadata mapping and vocabulary alignment

- American Fisheries Society 2025 WA-BC Chapter Annual Meeting workshop. 'Fishing for Clarity: Knowledge Modelling to Support Cross-organizational Collaboration and Data Sharing about Salmon Escapement

7.3 Support and follow through on Community Engaged Research (e.g. The Salmon Prize Project) that provides tangible value to the communities in which research or monitoring was conducted.

Appendix 2: Training Roadmap for Salmon Biologists Transitioning to Data Stewardship

This roadmap organizes training into three levels so biologists can build stewardship capacity in stages while applying skills directly to salmon data work.

Level	Suggested timeline	Primary goal
Beginner	0-3 months	Build core stewardship literacy and governance habits
Intermediate	3-9 months	Publish interoperable, citable salmon data products
Advanced	9+ months	Lead operational stewardship programs across teams

Beginner (0-3 months): Core stewardship literacy

Focus: establish a shared language for governance, metadata, and responsible data use.

- **Principles and core readings**
  - FAIR Data Principles
  - CARE Principles for Indigenous Data Governance
  - Wilkinson et al. 2016, FAIR Guiding Principles
  - Carroll et al. 2019, Indigenous Data Governance
- **Governance and planning tools**
  - DMPTool templates
  - DACI framework
- **Foundational skills training**
  - Data Carpentry Lessons
  - DataONE

Beginner SDIS starter tasks (salmon-domain):

- Read the Salmon Data Package specification.
- Treat the Salmon Domain Ontology as the canonical interoperability layer, then review the DFO Salmon Ontology as an organization-specific ontology and controlled vocabulary set aligned to that broader layer.
- Explore metasalmon documentation to understand package structure and metadata expectations.

Intermediate (3-9 months): Interoperable publication and integration

Focus: move from internal data handling to publishable, reusable data products.

- **Metadata and semantic standards**
  - ISO geospatial metadata overview (FGDC)
  - Ecological Metadata Language (EML)
  - Darwin Core
- **Persistent identifiers and credit**
  - DataCite
  - DataCite REST API intro
  - ORCID
  - ROR
- **Repository and API practice**
  - NPAFC data catalogue
  - PTAGIS API
  - NPAFC CKAN API example
  - Zenodo + GitHub citation guidance

Intermediate SDIS practicum tasks (salmon-domain):

- Build one versioned dataset package using metasalmon + the Salmon Data Package specification.
- Map at least 10 core fields to the Salmon Domain Ontology first, then use the DFO Salmon Ontology where organization-specific controlled vocabularies are needed.
- Publish a release with persistent metadata and DOI-backed citation text.

Advanced (9+ months): Operational stewardship leadership

Focus: institutionalize stewardship as an ongoing organizational capability.

- **Monitoring, incentives, and reporting**
  - COUNTER metrics
  - DataCite API
- **Community co-development and governance coordination**
  - RDA Salmon Research and Monitoring Interest Group
  - PNAMP/AFS "Fishing for Clarity" workshop example
  - User-centred design reference (Interaction Design Foundation)

Advanced SDIS capstone (salmon-domain):

- Operate an end-to-end stewardship cycle (governance, packaging, validation, publication, citation, and reuse monitoring) for a recurring salmon data product.
- Produce a quarterly stewardship report that summarizes metadata quality, release cadence, and reuse indicators.

This progression is intended to be practical rather than prescriptive: teams can adjust pace and tooling while keeping the same capability milestones.

Appendix 3: Getting Started Checklist

Use this practical checklist to assess how well your project, program, or organization aligns to the seven Best Practices. Start at the Project level, then scale to Program and Organization. Check off items you've completed and note gaps to prioritize.

Tip: For each item, capture a link to the living source (e.g., repository, shared drive, policy page) and the responsible owner.

Practice 1 — Make Data Governance Explicit

Project

```
- [ ] Do you have a Data Management Plan (DMP) covering scope, sensitive data, retention, and
- [ ] Is there a RACI (Responsible, Accountable, Consulted, Informed) table for key tasks? (
- [ ] Are Indigenous knoweledge holders or community members involved in the project?
- [ ] Are Indigenous Data Sovereignty (IDS) requirements identified and documented (who to co
- [ ] Is a data product charter written for each dataset or analysis product with purpose, au
```

Program

```
- [ ] Are DMP and charter templates standardized across projects and stored centrally?
- [ ] Are role definitions for Data Steward, Product Owner, and Maintainer explicit and assig
- [ ] Is this 'community-engaged' research that provides tangible benefit to communities?
- [ ] Are data sharing agreements/MOUs and ethical review pathways documented and reusable?
```

Organization

```
- [ ] Does a governance policy exist that sets minimum requirements for DMPs, RACI, retentior
- [ ] Is there a standing review forum (e.g., monthly data governance check-in) and a registr
```

Evidence to collect: DMP link, data product charter(s), RACI, IDS guidance, sharing agreements registry.

Practice 2 — Reuse Proven Infrastructure

Project

```
- [ ] Have you researched the existing data sharing infrastructure and data storage options s
- [ ] Is your code in version control (e.g., Git) with an issue tracker and releases?
- [ ] Are you using an approved repository or data store rather than creating a new silo? (wh
- [ ] Do you use existing organization authentication/authorization and backup processes?
```

Program

```
- [ ] Is there a preferred stack list (storage, metadata catalog, workflow runner, packaging
- [ ] Do projects consistently deposit finalized data in approved repositories with clear int
```

Organization

```
- [ ] Are enterprise services available and documented (data lake, object store, catalog/port
- [ ] Is there a deprecation pathway for legacy systems and a migration plan for priority dat
```

Evidence to collect: repository URLs, infrastructure inventory, intake criteria, backup/DR documentation.

Practice 3 — Use Persistent Identifiers (PIDs) for People, Projects, Data, and Methods

Project

```
- [ ] Do all contributors have ORCID IDs recorded in metadata?
- [ ] Does the project have a resolvable PID (e.g., DOI for a project page or protocol, inter
- [ ] Are datasets assigned DOIs (or other PIDs) at publication, and are versions tracked?
- [ ] Are methods/protocols published and citable (e.g., protocol DOI) and linked from datase
```

Program

```
- [ ] Is there guidance on when to mint PIDs, by whom, and where they resolve?
- [ ] Are projects linked to organizational identifiers (e.g., ROR for institutions) in metad
```

Organization

```
- [ ] Is there a PID policy and a provider/registrar configured (e.g., DataCite) with a docum
- [ ] Are PID linkages automated in the catalog (people  projects  datasets  publications)?
```

Evidence to collect: ORCID list, PID policy, DOI records, resolver links in the catalog.

Practice 4 — Shared Data Models, Vocabularies, and Metadata

Project

```
- [ ] Which metadata profile is used (e.g., ISO 19115, EML)? Is the minimum profile complete
- [ ] Are core entities modeled consistently (stock/population IDs, locations, temporal cover
- [ ] Are controlled vocabularies/code lists applied for key fields (e.g., species codes, gea
- [ ] Is a data dictionary included with definitions, units, allowed values, and provenance i
```

Program

```
- [ ] Do projects use a shared schema and code lists across datasets to enable easy joins?
- [ ] Are validation checks in CI (schema validation, vocab checks) standardized across repos
```

Organization

```
- [ ] Is there an endorsed salmon domain profile and shared code lists with owners and change
- [ ] Are mappings to external standards maintained (e.g., taxonomic, geospatial, hydrologica
```

Evidence to collect: metadata profiles, data dictionary, code lists, schema validators, mapping documentation.

Practice 5 — Store and Analyze Data for Easy Access, Use, and Trust

Project

```
- [ ] Is raw data immutable and separated from processed/analysis outputs?
- [ ] Is there a fully reproducible workflow (scripts/notebooks + environment + parameters) t
- [ ] Is the computational environment captured (lockfile/conda env, container image) and ver
- [ ] Are QA/QC checks automated with logs and thresholds documented?
- [ ] Are access controls and sensitive data handling documented and implemented?
```

Program

```
- [ ] Do projects follow a common repo layout and release process (tags, changelog, signed ar
- [ ] Are standard storage classes, lifecycle policies, and archival rules applied?
```

Organization

```
- [ ] Are security, backup/retention, and audit requirements defined and routinely verified?
- [ ] Is there a trusted long-term archive with fixity checking and preservation metadata?
```

Evidence to collect: workflow definition, environment files, container references, QA/QC reports, storage/backup settings.

Practice 6 — Incentivize and Track Sharing and Reuse

Project

```
- [ ] Is a clear citation and license statement included in metadata and README?
- [ ] Are reuse metrics collected (downloads, citations, API hits) and reviewed?
- [ ] Do release notes document what changed and implications for users?
```

Program

```
- [ ] Are common metrics dashboards available for priority datasets and updated automaticall
- [ ] Are data citations tracked in assessments, reports, and staff evaluations?
```

Organization

```
- [ ] Do policies require citation guidance and permissive, appropriate licensing where poss
- [ ] Are automated reports of reuse (e.g., via DOI provider APIs) delivered to product owne
```

Evidence to collect: LICENSE, CITATION, reuse dashboard link, policy excerpts, sample citations in reports.

Practice 7 — Build Community Through Co-Development and Mutual Benefit

Project

```
- [ ] Are stakeholders identified, including First Nations/Tribes/Indigenous partners, and e
- [ ] Have you held at least one co-design session to validate user needs and success criter
- [ ] Is there an open feedback channel (issues form, contact) and a published roadmap?
```

Program

```
- [ ] Do cross-project working groups exist for models, vocabularies, and tooling with regul
- [ ] Are community contributions recognized (authorship, acknowledgements, meeting time, fu
```

Organization

```
- [ ] Is there an endorsed governance body or community of practice with decision records?
- [ ] Are procurement/funding mechanisms available to support shared components and Indigeno
```

Evidence to collect: stakeholder map, engagement records, roadmap, working group notes, decision log.

Quick Start: 30/60/90-Day Plan

- First 30 days
  - ☐ Create/standardize DMP + RACI; draft data product charters for top 1–2 datasets.
  - ☐ Move code to version control; document repo structure; capture environment file.
  - ☐ Choose metadata profile and draft a minimal data dictionary; list code lists in use.
- By 60 days
  - ☐ Mint/plan PIDs (project page/protocols), add ORCIDs to metadata, prepare DOI for first dataset.
  - ☐ Add schema + vocab validation to CI; separate raw/processed; automate QA/QC checks.
  - ☐ Stand up a reuse dashboard or basic metrics capture; add citation/license to README and metadata.
- By 90 days
  - ☐ Publish first governed release to approved repository with DOI and complete metadata.
  - ☐ Formalize cross-project working group and change control for vocabularies.
  - ☐ Document archival/retention path and verify backups; schedule governance reviews.

Minimal Artifacts Checklist (Project Level)

- ☐ DMP (with IDS considerations) and RACI
- ☐ Data product charter(s) for priority dataset(s)
- ☐ Versioned repository with releases and changelog
- ☐ Metadata profile file + data dictionary + code lists
- ☐ Reproducible workflow + environment file or container
- ☐ QA/QC checks and results log
- ☐ Citation and license statements
- ☐ Plan for PIDs (ORCID list, dataset/protocol DOIs)
- ☐ Evidence of stakeholder engagement and roadmap

Maintain this list as a living issue in your repository and review quarterly.